

MA.912.LT.4 *Develop an understanding of the fundamentals of propositional logic, arguments and methods of proof.*

MA.912.LT.4.3

Benchmark

MA.912.LT.4.3 Identify and accurately interpret “if...then,” “if and only if,” “all” and “not” statements. Find the converse, inverse and contrapositive of a statement.

Benchmark Clarifications:

Clarification 1: Instruction focuses on recognizing the relationships between an “if...then” statement and the converse, inverse and contrapositive of that statement.

Clarification 2: Within the Geometry course, instruction focuses on the connection to proofs within the course.

Connecting Benchmarks/Horizontal Alignment

- MA.912.GR.1
- MA.912.GR.2.6, MA.912.GR.2.7, MA.912.GR.2.8, MA.912.GR.2.9
- MA.912.GR.5

Terms from the K-12 Glossary

Vertical Alignment

Previous Benchmarks

- MA.8.GR.1

Next Benchmarks

- MA.912.LT.4
- MA.912.LT.5

Purpose and Instructional Strategies

In grade 8, students were first introduced to theorems and their converses with the Triangle Sum, Triangle Inequality and Pythagorean Theorems and used, informally, logical statements about the angles formed by crossing lines and angles in polygons. In Geometry, students use and think more formally about different kinds of logical statements. In later courses, students refine their knowledge and skills regarding logic and set theory.

- Instruction of this benchmark should be done throughout the course, as students are using postulates, proving relationships and theorems and studying definitions.
- Instruction includes the student understanding that a good definition is precise and accurate, and can always be written in the “if and only if” form. Having this understanding helps students to avoid misconceptions within definitions.
 - For example, the definition of a parallelogram states that it is a quadrilateral with opposite sides parallel. This can be written as “a quadrilateral is a parallelogram if and only if its opposite sides are parallel.” This means that “if a quadrilateral is a parallelogram, then it has opposite sides that are parallel,” and that “if a quadrilateral has opposite sides that are parallel, then it is a parallelogram.”
- Instruction includes discussing the meaning of “all” within statements.
 - For example, students can discuss whether it is true that *all* translations are rigid motions or whether it is true that *all* rigid motions are translations.
- Instruction includes the student understanding that a conditional statement has a

hypothesis and a conclusion in the form of “if...then,” where the *if* part of the statement is the hypothesis and the *then* part of the statement is the conclusion. A conditional statement can be transformed by negating or rearranging its parts to create an inverse statement, a converse statement, a contrapositive statement and an “if and only if” statement.

- Conditional Statement
When a statement is in form of “if...then,” where the *if* part of the statement is the hypothesis and the *then* part of the statement is the conclusion.
For example, given the conditional statement “If two angles are vertical, then the angles are congruent,” the hypothesis is “two angles are vertical” and the conclusion is “the angles are congruent.”
- Negation
When a statement uses the word “not.”
For example, “two angles are not vertical” is the negation of the hypothesis and “the angles are not congruent” is the negation of the conclusion.
- Inverse Statement
When both the hypothesis and conclusion of a conditional statement are negated.
For example, “If two angles are not vertical, then the angles are not congruent.”
- Converse Statement
When the hypothesis and the conclusion are switched.
For example, “If two angles are congruent, then the angles are vertical.”
- Contrapositive Statement
When both the hypothesis and conclusion of a conditional statement are negated and switched.
For example, “If two angles are not congruent, then the angles are not vertical.”
- “If and only if” Statement
When a conditional statement and its converse are combined in an abbreviated way.
For example, “Two angles are congruent, if and only if the two angles are vertical,” is a compact way of saying “If two angles are vertical, then the angles are congruent, and if two angles are congruent, then the angles are vertical.”
- Instruction includes interpreting conditional statements and negations or rearrangements of their hypothesis and conclusion and analyzing whether the resulting statements are true or false. Students can use postulates, definitions and proofs to determine whether a statement is true. Students can use counterexamples to show that a statement is false.
 - Given a true conditional statement, its converse statement or inverse statement may or may not be true.
For example, given the true conditional statement “If an angle measure is 34° , then the angle is acute,” its converse or inverse would not produce a true statement.
For example, given the true conditional statement “If a triangle has side lengths a , b and c satisfying $a^2 + b^2 = c^2$, then the triangle is a right triangle,” its converse would produce a true statement.
 - Given a true conditional statement, its contrapositive statement is also true.
For example, given the true conditional statement “If an angle measure is 34° , then the angle is acute,” its contrapositive would produce a true statement.

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- For enrichment of this benchmark, instruction includes the use of truth tables to help students organize their work.

Common Misconceptions or Errors

- Students may confuse the hypothesis and conclusion.
- Students may try to change the inverse, converse or contrapositive to make sense in the real world rather than using the logic definitions.

Instructional Tasks

Instructional Task 1 (MTR.7.1)

Use the statements below to identify the converse, inverse and contrapositive of the statement “If I can run a 5K race in under 27 minutes, then I can start the race at the front of the pack.”

- If I cannot run a 5K race in under 27 minutes, then I can start the race at the front of the pack.
- If I cannot run a 5K race in under 27 minutes, then I cannot start the race at the front of the pack.
- If I can run a 5K race in under 27 minutes, then I cannot start the race at the front of the pack.
- If I cannot start the race at the front of the pack, then I cannot run a 5K race in under 27 minutes.
- If I can start the race at the front of the pack, then I can run a 5K race in under 27 minutes.

Instructional Task 2 (MTR.4.1)

Part A. Write an “if...then” statement involving a quadrilateral.

Part B. Rewrite the statement as an “if and only if” statement. How are the two statements different in their meaning?

Instructional Items

Instructional Item 1

Use the following statement to answer the questions.

A triangle is an equilateral triangle if and only if the triangle has three congruent sides.

Part A. Write the two “if...then” statements that can be written from the given statement.

Part B. Write the converse of one of the conditional statements created in Part A.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*